

Longitudinal Gaps and Readout Connectivity in a Pilot Generative Model of Zero-Degree Calorimeter Showers

Julian Juan

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Abstract

We study a pilot generative model of single-neutron Geant4 showers in a 6,790-channel Zero-Degree Calorimeter readout. Conditioned on neutron energy and direction, the model samples a total readout deposit, active layers, layer energies, active channels, and channel energies. We compare one checkpoint trained on 26,624 events with reference showers at 10,000 stored conditions between 50 and 250 GeV incident kinetic energy. The mean total deposit differs by 0.045 GeV (1.0%), partly because opposite electromagnetic and hadronic shifts cancel. Among nonempty events, mean active-layer counts differ by 0.25, but the generated showers reach 4.42 layers farther downstream and contain 3.95 more inactive layers within their occupied span. Mean active-channel counts differ by -28.7 (-1.8%); mean component counts on the model graph differ by $+35.97$. Because that graph links only adjacent layers, inactive layers can themselves split the support. The comparison is descriptive for one checkpoint on a repeatedly inspected validation sample. The retained aggregates cannot separate longitudinal splitting from within-layer fragmentation or establish the structural difference’s uncertainty, threshold dependence, or persistence across fits.

1 Introduction

Detailed particle transport is central to detector studies, but calorimeter shower simulation is computationally expensive. Generative models based on normalizing flows, diffusion, point clouds, graph networks, and conditional flow matching have been developed for faster sampling [1–5]. CaloChallenge evaluates such models with shower observables, distributional distances, classifier tests, model size, and generation time [6]. Different tests can reveal different distributional failures [7].

Zero-Degree Calorimeters (ZDCs) have their own fast-simulation literature. Studies of the ALICE ZDC have used variational autoencoders, generative adversarial networks, diffusion models, conditional flow matching, and mixtures of generative experts [8–12]. A recent normalizing-flow study also tests co-activation of channels [13]. Those studies use a different detector representation, so their numerical results cannot be compared directly with the raw deposited-energy vector examined here.

Spatial organization matters in calorimetry. ATLAS, for example, clusters neighboring cells above signal-significance thresholds and uses cluster shapes in reconstruction [14]. Here we examine connectivity on a different, model-defined graph with a zero threshold. Its components are diagnostic properties of the stored readout, not reconstructed shower clusters.

We examine one pilot-trained checkpoint on an irregular 65-layer readout. The 10,000-condition comparison was inspected repeatedly during development, and only its aggregate report is available in the public package. We compare total and section deposits, longitudinal activity, channel occupancy, and graph components to locate disagreements that a total-response mean would hide.

2 Simulation target and evaluation population

For each event, the stored condition is an incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$. The model input is

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \mathbf{p}/\|\mathbf{p}\|, \quad (1)$$

where $K_{\text{inc}} = E - m_n$. The two energy features are deterministically related by $E = K_{\text{inc}} + m_n$; retaining both is a frozen implementation choice and does not add independent conditioning information.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ contains raw, nondigitized energy deposits summed by valid readout ID from a Geant4 simulation [15]. It is neither the incident-neutron energy nor all energy lost in detector material. Layer 0 contains 400 electromagnetic-calorimeter (ECAL) channels. Hadronic-calorimeter (HCAL) layers 1–63 contain 100 channels each and layer 64 contains 90, for 6,390 HCAL channels. Of the HCAL readout IDs, 2,400 gang two to four stable physical positions. Each ganged channel is represented by the unweighted centroid of its positions (Fig. 1). The released metadata do not specify the full material composition, Geant4 version, physics list, production cuts, field configuration, or whether p^μ is recorded at generation or calorimeter entry. The few-GeV readout totals are therefore reported as deposits in this stored target, without interpreting their ratio to incident energy as a calibrated response.

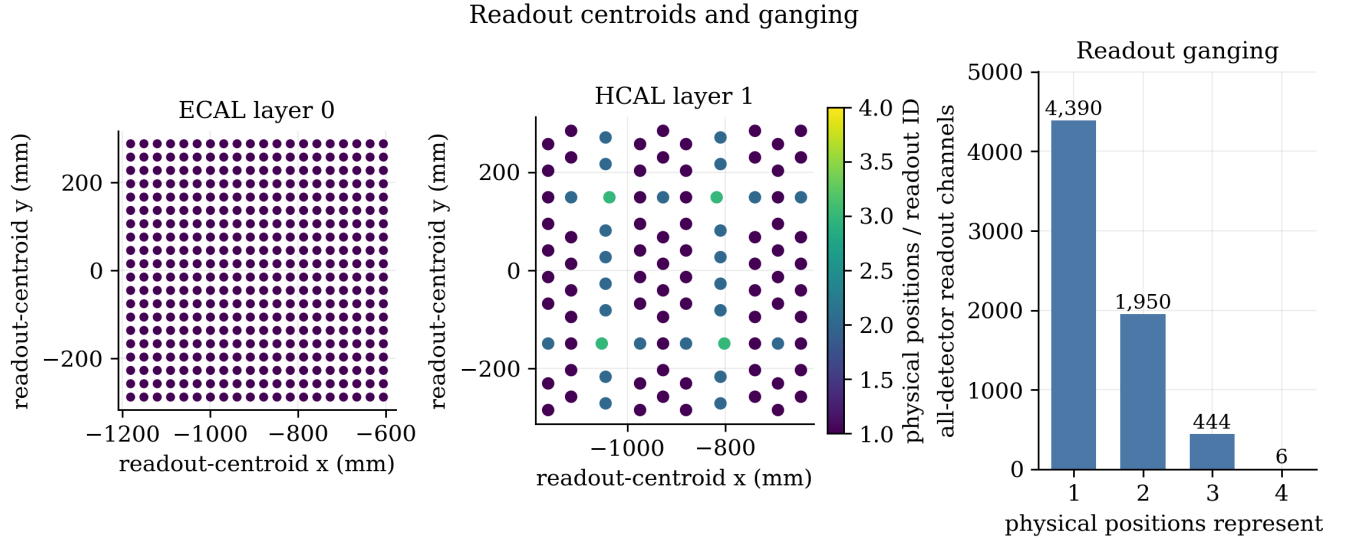


Figure 1: Frozen readout geometry. The left panels show ECAL layer 0 and HCAL layer 1. The ganging histogram covers all 6,790 readout channels; its first bar contains 400 ECAL and 3,990 HCAL channels.

The graph connects every channel to its eight nearest same-layer centroids. For each node in layer $\ell = 0, \dots, 63$, four nearest centroids are selected in layer $\ell + 1$, and each selected longitudinal pair is stored in both directions. This gives $8 \times 6790 = 54,320$ lateral directed edges and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal directed edges, or 107,920 in total. The same graph is used by the generator and by the connectivity diagnostic. It is an implementation graph, not a uniquely established physical-neighbor graph, and centroiding removes separation within a ganged channel.

The preparation record contains 764,940 events over 0–300 GeV, divided by deterministic event hash into 612,482 training, 76,158 validation, and 76,300 nominal test events. The checkpoint studied here used a smaller pilot of 26,624 training events and 6,656 validation events. Its training population is 4.35% of the canonical training partition.

All shower comparisons use a 10,000-condition sample from the canonical validation partition with $50 \leq K_{\text{inc}} \leq 250 \text{ GeV}$. We call this sample D_{dev} . It was inspected during model and manuscript development, and its overlap with the 6,656-event pilot-validation sample is not recoverable from the released aggregates. The pilot and canonical split manifests have different hashes because they describe different partitions; the recorded preparation cross-tabulation places pilot training events within canonical training and pilot validation events within canonical validation. No nominal test event is used in this manuscript. The stored comparisons are development-sample measurements, not an untouched test of expected performance.

3 Generator and training procedure

The generator has 2,082,507 trainable parameters. At generation it receives only the five features in Eq. (1). A two-block residual multilayer perceptron maps them to a 128-component representation h_c , which conditions each sampling stage. The discrete stages predict probability distributions for visibility, layer activity, and channel counts or selection. Two conditional flows transform independent Gaussian draws into continuous layer-energy and channel-energy allocations. The model has no learned shower encoder or common random latent vector: independent draws enter at the discrete heads and at the two flows.

Figure 2 follows the sampling path. Each draw conditions later stages. Thus a change in total deposit can affect every later allocation, whereas the final channel-energy assignment cannot change a previously sampled layer budget or channel count.

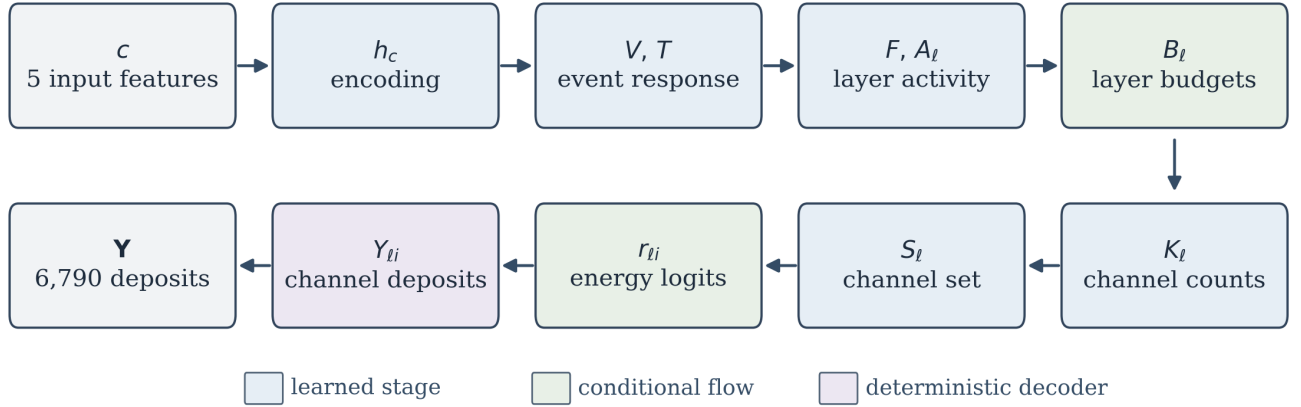


Figure 2: Event generation runs left to right across the top row, then right to left across the bottom row. Blue boxes are learned conditioning and sampling stages, green boxes are conditional flows, and the purple box is the deterministic decoder. V indicates any positive readout deposit, T is its total, F is the first active layer, A_ℓ is layer activity, B_ℓ is the layer budget, K_ℓ is the active-channel count, and S_ℓ is the selected channel set. The flow output $r_{\ell i}$ gives channel-energy logits, normalized within each layer to obtain $Y_{\ell i}$. During training, stages receive upstream quantities derived from the reference shower; generation uses earlier sampled outputs.

3.1 Total deposit and longitudinal allocation

A Bernoulli head samples V , which indicates whether the event has any positive readout deposit. A second head outputs the weights, means, and positive widths of a four-component Gaussian mixture for the transformed total deposit

$$q_T = \log(1 + T/10 \text{ GeV}). \quad (2)$$

When $V = 1$, the implementation samples a mixture component and a normal variate, applies the inverse transform, and sets a negative result to zero. It then limits the result to

$$0 \leq T \leq C(K_{\text{inc}}), \quad C(K_{\text{inc}}) = \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}. \quad (3)$$

The cap is a numerical bound derived from the training sample. If $V = 0$, or if inversion and clipping give $T = 0$, no layer is activated. The aggregate evaluation did not retain cap-activation or reference-exceedance counts, so the response tail near this boundary is untested.

For $V = 1$, a categorical head samples the first active layer F from (h_c, T) . Layers before F are set inactive, layer F is forced active, and a Bernoulli head samples the later activity indicators. The later indicators are independent conditional on the variables supplied to that head,

$$p(\mathbf{A} \mid h_c, T, F) = \prod_{\ell > F}^{64} p(A_\ell \mid h_c, T, F, \ell), \quad A_F = 1. \quad (4)$$

This head does not require the active layers to be contiguous.

The first conditional flow apportions T among active layers. For training, each reference layer deposit is divided by the reference total; the logarithms of these fractions, floored at 10^{-8} , are centered over active layers to form a target vector x_1 . The flow draws a standard-normal vector x_0 and a time t uniformly in $[0, 1]$, then constructs

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{CFM}} = \|v_\theta(x_t, t; \text{context}) - (x_1 - x_0)\|_2^2, \quad (5)$$

The squared error trains the velocity field v_θ to point along the straight path from x_0 to x_1 [16]. At generation, the field advances a fresh Gaussian vector through eight fixed updates, with velocity evaluated at the midpoint time of each interval. A softmax over active layers converts its output to nonnegative fractions π_ℓ that sum to one. The layer budgets are

$$B_\ell = T\pi_\ell, \quad B_\ell = 0 \text{ when } A_\ell = 0, \quad \sum_{\ell=0}^{64} B_\ell = T. \quad (6)$$

The velocity field uses a 128-component layer representation and two bidirectional transformer layers with four attention heads. Its context comprises h_c , T , the sampled activity mask, layer identity, and a sinusoidal encoding of flow time.

3.2 Channel count, support, and deposited energy

For each layer, a categorical head receives (h_c, B_ℓ, A_ℓ) and a learned layer identifier, then samples the active-channel count K_ℓ . Its feasibility mask permits $K_\ell = 0$ in an inactive layer and $1 \leq K_\ell \leq N_\ell$ in an active layer, where N_ℓ is that layer’s channel count. The mask fixes a count but imposes no minimum energy per channel in this zero-threshold study.

The support network scores every channel for selection. Its inputs for each channel are standardized (x, y, z) coordinates, normalized layer index, ECAL/HCAL indicators, h_c , the layer budget B_ℓ , and the requested occupancy fraction K_ℓ/N_ℓ . Three 96-component graph blocks update each channel representation from its neighbors. Their five edge inputs specify coordinate differences, distance, and whether an edge is lateral, forward-longitudinal, or backward-longitudinal. Channel representations are averaged within each layer; a two-layer, four-head bidirectional transformer exchanges information among the 65 layer summaries; the resulting context is returned to each channel. Independent Gumbel noise is added to the channel scores, and the K_ℓ highest scores select exactly K_ℓ channels in layer ℓ [17]. Selection has no constraint that would make those channels a connected set.

The second conditional flow divides each sampled layer budget among its selected channels. Its training target is obtained by taking the log of each reference channel’s fraction of its layer energy, flooring fractions

at 10^{-8} , and centering the logs within that layer. It uses the same straight-path objective as Eq. (5). Its velocity network has the support network’s graph and layer-context structure, with the current flow state and a sinusoidal time encoding added at each channel. Unselected channels are masked before and after every graph block. At generation, independent Gaussian values on selected channels undergo eight fixed updates to produce logits $r_{\ell i}$. The deterministic decoder computes

$$Y_{\ell i} = \begin{cases} B_{\ell} \frac{\exp r_{\ell i}}{\sum_{j \in S_{\ell}} \exp r_{\ell j}}, & i \in S_{\ell}, \\ 0, & i \notin S_{\ell}, \end{cases} \quad (7)$$

where S_{ℓ} is the selected set. Thus, in exact arithmetic, $|S_{\ell}| = K_{\ell}$, $Y_{\ell i} \geq 0$, $\sum_i Y_{\ell i} = B_{\ell}$, and $\sum_{\ell i} Y_{\ell i} = T$. These identities conserve the model’s sampled readout budget. They do not enforce conservation of incident-neutron energy, agreement with Geant4, or connectivity of the selected support.

3.3 Training and checkpoint selection

Training derives V , T , F , A_{ℓ} , B_{ℓ} , K_{ℓ} , S_{ℓ} , and both flow targets from each reference shower. A component receives the reference values of earlier components, whereas generation passes its own sampled values downstream. This training procedure is often called teacher forcing. The nine losses are binary cross entropy for visibility and activity, mixture negative log likelihood for T , categorical cross entropy for F and K_{ℓ} , binary cross entropy plus pairwise ranking for support, and mean-squared velocity error for each flow. Weights were obtained from gradient norms measured on training data. The response, layer-profile, and count weights reached the lower clipping bound, and the visibility weight reached the upper bound. These weights describe the optimization; they do not identify the source of the connectivity discrepancy.

The selected joint-training run used AdamW with a peak learning rate of 3×10^{-4} , batch size six, accumulation over four batches, gradient clipping at one, and seed 20260723. Epoch 90 has the lowest recorded validation objective among the 104 retained rows spanning epochs 11–114 and defines the checkpoint studied here. Validation objectives are lower than training objectives over much of this record. The retained aggregates do not identify why, so the ordering is not given a physical interpretation. Both flows use eight integration updates at evaluation; no solver-step comparison was retained.

4 Diagnostics

We compare mean total, ECAL, and HCAL deposits over all events. For eight 25 GeV incident-energy bins, the report retains means and standard deviations but no binwise uncertainty estimates. A paired, incident-energy-stratified bootstrap with 1,000 resamples supplies a 95% percentile interval for the difference in zero-deposit fractions. It does not supply intervals for the structural metrics.

A layer or channel is active when its stored energy is strictly positive; no detector or analysis threshold is applied. For a nonempty event with first and last active layers F and L , the occupied span is $L - F + 1$, and the number of inactive layers inside it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_{\ell}. \quad (8)$$

This counts inactive layers, not contiguous runs. A shower has an interior gap when $G > 0$. Ignoring edge direction, the occupied channels S form connected components $C_1, \dots, C_m$ on the model graph. We report their number m and the fraction of occupied channels in the largest one, $\max_j |C_j|/|S|$. These support summaries use only nonempty events: 9,907 reference and 9,858 generated events. Because longitudinal graph edges connect adjacent layers only, a contiguous run of inactive layers between occupied layers necessarily separates the support on either side. The retained value of G cannot tell us how many such runs occur.

The retained condition-only pipeline control has AUROC 0.500. It contains no shower variables and checks only the pairing of identical condition distributions across labels. The original battery also records shower-aware classifier scores, but its row-wise random train/test split can place members of a matched pair on opposite sides of the split. We exclude those scores rather than interpret them as shower-fidelity tests. A future classifier test should hold out matched pairs together [7, 18].

5 Results

Table 1 reports absolute differences so that ratios on small denominators do not exaggerate the comparison. The all-event mean total deposit differs by 0.045 GeV, but this sum combines an ECAL excess of 0.175 GeV with an HCAL deficit of 0.129 GeV. Across the eight incident-energy bins, the generated-minus-reference mean difference ranges from -7.7% to $+7.7\%$, while the standard-deviation difference ranges from -10.9% to $+9.6\%$. Without binwise uncertainties these ranges diagnose possible cancellation but do not establish a statistically resolved energy dependence.

Table 1: Aggregate development-bank results. Deposit rows and zero-deposit counts use all 10,000 pairs. Support rows are conditional on nonempty events (9,907 reference and 9,858 generated). Percentage-point differences are denoted pp. The report supplies a bootstrap interval for the zero-deposit fraction difference, but no structural intervals.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

The generated shower reaches farther along the layer axis despite having nearly the same number of active layers. Its mean first active layer is 0.225 layer later and its mean last active layer 4.42 layers later. The occupied span grows by 4.20 layers while the active-layer count grows by only 0.25, leaving 3.95 more inactive layers inside the span on average. Figure 3 shows the mean energy allocation.

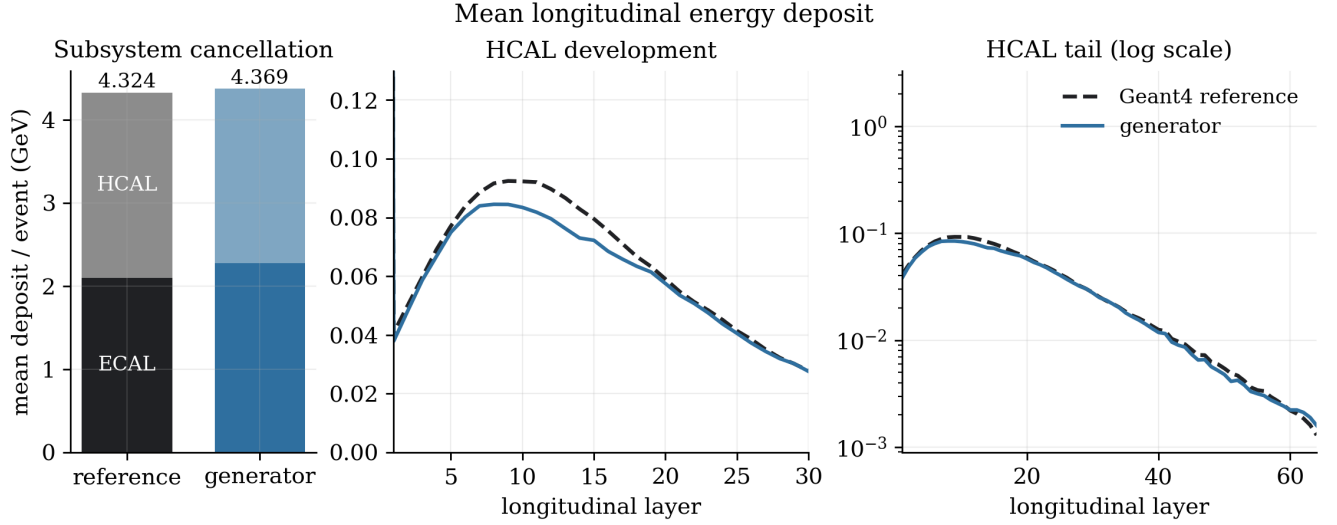


Figure 3: Mean longitudinal allocation on D_{dev} . The stacked bars show the opposing ECAL and HCAL shifts. The layer panels show HCAL layers only; ECAL layer 0 is included in the bars. Averaging over events hides individual interior gaps; no uncertainty bands were retained.

At channel level, the generated sample has 28.7 fewer active channels and 35.97 more weak components on average. Its mean largest-component fraction is lower by 6.03 percentage points. These component statistics describe the same strict-positive support that contains the additional longitudinal gaps. The aggregate report does not distinguish components separated by empty layers from those arising within occupied layers, nor does it record component energies. A small positive deposit counts as occupied under this definition even if it would be removed by a realistic threshold.

The mean fraction of directed graph edges with both endpoints occupied is 0.1375 for generated showers and 0.1458 for the reference, averaged over all 10,000 events. This local co-occupancy statistic is subject to the same graph and threshold choices as the component counts.

There are 49 more zero-deposit generated events than reference events, a difference of 0.49 percentage points. The stored paired bootstrap gives a nominal 95% percentile interval of 0.19–0.76 percentage points for this difference. Because the same validation sample was repeatedly inspected, this interval describes resampling variation within the selected bank; it is not a confirmatory coverage statement for a new sample. Empty events are excluded from the support statistics and therefore change their denominators.

The decoder-level checks passed for all 1,250 batches of eight generated events. No nonfinite value, negative value, invalid support, requested-versus-realized count mismatch, or positive-support mismatch was recorded. Maximum event and layer budget residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV under the evaluator’s batch-wide absolute-plus-relative tolerance. These checks establish consistency with sampled budgets and counts for this generated bank; they do not test agreement with Geant4.

6 Discussion

The longitudinal summaries show that generated showers extend farther downstream and contain more inactive layers between their first and last deposits. The graph-component difference is partly entangled with that result by construction. A contiguous empty-layer run separates occupied layers on an adjacent-layer graph, while additional components may also occur within occupied layers. The retained aggregates count empty layers but neither their runs nor the locations and energies of individual components. They therefore do not establish an independent lateral-fragmentation effect.

The model enforces layer budgets and active-channel counts, not adjacency among selected channels. Its independent later-layer activity draws and Gumbel-perturbed top- K_ℓ support selection could permit gaps and fragmented support. This is a mechanism the architecture allows, not an explanation isolated by the present comparison. Pilot training size, validation-objective checkpoint selection, and eight-step flow integration also remain untested contributors. A matched ablation or an oracle cascade that replaces one generated upstream variable at a time with its reference value would locate the first stage at which the discrepancy appears.

The strict-positive support includes arbitrarily small stored deposits; it is not a thresholded detector observable. Recomputing component energies and connectivity under physically motivated energy thresholds, alternative adjacency graphs, and fixed energy and direction slices is necessary before interpreting the difference as shower morphology.

D_{dev} was repeatedly inspected, its overlap with pilot validation is unresolved, and the public package contains aggregates rather than paired event arrays. It cannot provide structural uncertainty intervals or the threshold and graph scans above. Repeated generation of a fixed checkpoint would measure sampling variation; nearby checkpoints and at least three independently trained final seeds would test fit variation. A locked bank, complete simulation provenance, reconstructed-observable comparisons, and end-to-end timing would be needed to assess use as fast simulation. These measurements describe the stored checkpoint and bank.

7 Conclusion

For this pilot checkpoint, similar mean occupancies accompany a later, more interrupted longitudinal support in the stored validation comparison: mean active layers differ by +0.25, mean last active layer by +4.42, and mean inactive layers within the occupied span by +3.95. The generated sample has 35.97 more weak components on the model graph, but the adjacent-layer graph makes this statistic inseparable from longitudinal gaps without event-level analysis. The mean total deposit differs by only 0.045 GeV because the ECAL and HCAL shifts partly cancel. These findings identify where the present surrogate disagrees with its reference sample; they do not validate a fast-simulation replacement or identify a causal model defect.

Code and data availability

The manuscript source, aggregate validation report, event-independent geometry summary, training-history extract, figure builder, and QA records are available at <https://github.com/JulianAttemptsCoding/fast-mc-paper>. The collaboration-owned event file and checkpoint are not redistributed. The public package reproduces the manuscript and aggregate arithmetic; it cannot retrain the model, regenerate showers, or reconstruct event-level tests. Hashes identify the implementation files inspected for this paper, but the available record does not establish that these were the exact source bytes used during training.

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Author contributions

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

Use of generative AI

OpenAI Codex with GPT-5.6-Sol assisted with code and manuscript drafting, analysis and model-design proposals, literature searches, and editing. The author supplied the main research ideas, checked the resulting work, and is responsible for the scientific content and final manuscript.

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